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Entangling Unravellings for Quantum State Engineering and Quantum Metrology

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A coloro che mi hanno insegnato a non mollare mai

True happiness is to enjoy the present, without anxious dependence upon the future, not to amuse ourselves with either hopes or fears but to rest satisfied with what we have, which is sufficient, for he that is so wants nothing. The greatest blessings of mankind are within us and within our reach. A wise man is content with his lot, whatever it may be, without wishing for what he has not.
- Seneca

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Introduction

Continuous quantum measurement has emerged as a powerful route to engineer non-classical states and to extract metrologically useful information from noisy platforms. Rather than treating measurement backaction as a nuisance, one can tailor the measurement channel and its unravelling to steer a system toward entanglement and spin squeezing. This thesis explores that idea in the dephasing-dominated regime, focusing on *entangling unravellings* that convert which-path erasure into a resource for quantum state engineering and quantum metrology.

We study a minimal, experimentally motivated setting: two qubits that radiate into modes subsequently mixed on a balanced beam splitter and measured with different continuous readouts. The only decoherence is single-qubit dephasing (local Pauli- z noise). As a concrete baseline we initialize the system in the separable state $|++\rangle$ and then generalize to coherent spin states (CSS) at varying polar angles. We compare three measurement schemes: photodetection (quantum jumps), homodyne (single-quadrature), and heterodyne (double-quadrature) which realize distinct stochastic unravellings of the same master equation and probe symmetric/antisymmetric combinations of the local dephasing channels. In doing so, we highlight the information backaction trade-off between homodyne and heterodyne detection and show how interferometric mixing can erase which-path information and promote conditional entanglement.

Methodologically, we formulate the stochastic master equation (SME) with an explicit split between monitored and unmonitored channels, allowing us to model finite detection efficiency as a single parameter without changing the underlying system dynamics. Analytical results are derived in the two-qubit case: we obtain closed-form jump probabilities and conditional states for the interferometric photodetection channels, and we provide compact expressions and bounds for trajectory-averaged entanglement and spin squeezing. These predictions are validated against numerical quantum-trajectory simulations implemented in QuTiP, ensuring internal consistency across jump and diffusive unravellings and quantifying robustness to inefficiency.

From the metrological perspective, we track standard figures of merit such as concurrence and the Kitagawa-Ueda and Wineland spin-squeezing parameters, always evaluating them in the instantaneous mean-spin frame to capture the rotation of the optimal squeezing axis induced by measurement backaction. Moving beyond the specific $|++\rangle$ input, we map how the entanglement yield and squeezing depth depend on the CSS orientation and on the choice of readout. This reveals simple design rules for maximizing performance under dephasing: the alignment of the input state relative to the measured quadrature, the interferometric phase that selects symmetric versus antisymmetric channels, and minimum efficiency thresholds required to overcome information loss.

Finally, we outline a scalable interferometric architecture for atomic ensembles that implements a collective J_z measurement together with complementary observables accessible via alternative unravellings. We extend our analysis beyond two qubits to assess concurrence and conditional squeezing in larger systems, discussing scaling with the number of atoms and the role of measurement inefficiency in the many-body limit. The overall picture is that, even with pure dephasing as the dominant noise, appropriately

engineered continuous measurements can serve as a practical and platform-agnostic resource for quantum state preparation and precision sensing.

Contributions.

- A unified, interferometric framework to compare photodetection, homodyne, and heterodyne *entangling unravellings* under single-qubit dephasing.
- Closed-form analytics for two qubits (jump probabilities, conditional states) with quantitative, trajectory-averaged benchmarks for concurrence and spin squeezing.
- An explicit efficiency model via monitored/unmonitored channel splitting in the SME, with validated thresholds from QuTiP simulations.
- Generalization from the $|++\rangle$ input to CSS with varying polar angle, identifying orientation-dependent optima and readout-specific trade-offs.
- Extension to $N > 2$ qubits and a scalable readout that realizes collective J_z together with complementary observables relevant for metrology.

Structure of the thesis. Chapter 1 reviews continuous measurement theory and stochastic unravellings. Chapter 2 introduces the two-qubit interferometric model and derives analytic results under dephasing. Chapter 3 presents numerical validations and the treatment of finite efficiency. Chapter 4 generalizes to CSS inputs and discusses metrological performance. Chapter 5 extends to multi-qubit ensembles and outlines scalable architectures and applications.

1

Two qubit system: looking for entanglement

1.1 Convention, states and master equation

We consider a system of N qubits (with particular emphasis on the $N = 2$ case), each described in the computational basis $\{|0\rangle, |1\rangle\}$. Throughout this work we adopt the convention

$$\sigma_z |1\rangle = +|1\rangle, \quad \sigma_z |0\rangle = -|0\rangle, \quad (1.1)$$

so that the excited state $|1\rangle$ corresponds to the positive eigenvalue of σ_z . This choice fixes all relative signs in the collective spin operators and in the dissipative dynamics. The single-qubit spin operators are defined as

$$S_\alpha^{(i)} = \frac{1}{2} \sigma_\alpha^{(i)}, \quad \alpha \in \{x, y, z\}, \quad (1.2)$$

where the superscript i labels the qubit. Collective spin operators are introduced as

$$J_\alpha = \sum_{i=1}^N S_\alpha^{(i)}. \quad (1.3)$$

When needed, spin coherent states (CSS) are written as

$$|\theta, \phi\rangle = \bigotimes_{i=1}^N \left(e^{i\phi} \sin \frac{\theta}{2} |1\rangle_i + \cos \frac{\theta}{2} |0\rangle_i \right), \quad (1.4)$$

which corresponds to a product state with all spins pointing along the direction $\hat{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ on the Bloch sphere. In particular, the eigenstates of σ_x are

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \quad (1.5)$$

We use the standard Bell states

$$|\phi_\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}} \quad (\text{even } z\text{-parity}), \quad (1.6)$$

$$|\psi_\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}} \quad (\text{odd } z\text{-parity}). \quad (1.7)$$

The dynamics of the system is described within the Markovian open-quantum-system framework. Under the standard assumptions of weak system–environment coupling,

short bath correlation time (Born–Markov approximation), and complete positivity, the reduced density operator $\rho(t)$ obeys a Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation,

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_k \mathcal{D}[L_k]\rho^{(c)} dt = -i[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right). \quad (1.8)$$

where $\mathcal{D}[A]\rho = A\rho A^\dagger - \frac{1}{2}\{A^\dagger A, \rho\}$. Here H is the system Hamiltonian, and $\{L_k\}$ are the Lindblad or collapse operators describing dissipative processes and continuous measurements.

Equation (1.8) generates a completely positive and trace-preserving (CPTP) dynamical semigroup.

In this work we focus on dissipative channels associated with spin components along the z direction. For two qubits, a central role is played by collective jump operators of the form

$$\hat{L}_\pm = \sqrt{\gamma} \frac{\sigma_z^{(1)} \pm \sigma_z^{(2)}}{\sqrt{2}} \quad (1.9)$$

where γ sets the overall dissipation or measurement rate.

These operators naturally arise in measurement schemes where emissions from the two qubits are interfered (e.g. via a beam splitter) before detection, such that the measurement does not reveal which-path information. The two channels correspond to symmetric and antisymmetric combinations of local z -spin operators. While the master equation (1.8) uniquely determines the evolution of the density operator, its stochastic unravelling into quantum trajectories is not unique (see Appendix A). Different measurement schemes (photodetection, homodyne, heterodyne) correspond to different stochastic evolutions at the level of pure states, all reproducing the same ensemble-averaged dynamics.

In the photodetection unravelling associated with the jump operators (1.9), the system undergoes non-unitary no-click evolution interrupted by quantum jumps. Each jump corresponds to the application of $L_\pm$ to the state vector, followed by normalization, and reveals information about collective spin observables and parity.

This measurement backaction is the key mechanism exploited in this work for the generation and stabilization of entanglement and for the study of metrological quantities such as concurrence and spin squeezing.

1.2 Photodetection

A Markovian master equation admits a stochastic *unravelling* in terms of measurement-conditioned quantum trajectories. In the case of photodetection (quantum-jump monitoring), the measurement record is a set of counting processes $\{N_k(t)\}_k$ associated with the monitored output channels. In an infinitesimal time step dt , the increments

$$dN_k(t) \in \{0, 1\} \quad (1.10)$$

represent, respectively, a *no-click* event or a *click* in detector k . They are Poissonian random variables with conditional mean

$$\mathbb{E}[dN_k(t)] = \eta \text{Tr} \left[L_k^\dagger L_k \rho^{(c)}(t) \right] dt, \quad (1.11)$$

where $\eta \in [0, 1]$ is the detection efficiency and $\rho^{(c)}(t)$ denotes the conditional (normalized) system state.

For jump monitoring, the conditional evolution can be written as a piecewise deterministic process: a smooth non-unitary evolution conditioned on $dN_k = 0$, interrupted by stochastic jumps when $dN_k = 1$. A convenient Itô form of the stochastic master equation (SME) is

$$\begin{aligned} d\rho^{(c)} = & -i[H, \rho^{(c)}] dt + (1 - \eta) \sum_k \mathcal{D}[L_k] \rho^{(c)} dt \\ & + \sum_k \left(\frac{L_k \rho^{(c)} L_k^\dagger}{\text{Tr}[L_k^\dagger L_k \rho^{(c)}]} - \rho^{(c)} \right) dN_k(t), \end{aligned} \quad (1.12)$$

By averaging Eq. (1.12) over the measurement outcomes one recovers the unconditional GKSL master equation.

The resulting photodetection SME reads

$$d\varrho_t^{(c)} = -i\omega[J_z, \varrho_t^{(c)}]dt + (1 - \eta) \frac{\gamma}{2} \sum_j \mathcal{D}[\sigma_z^{(j)}] \varrho_t^{(c)} dt + \sum_j \left(\sigma_z^{(j)} \varrho_t^{(c)} \sigma_z^{(j)} - \varrho_t^{(c)} \right) dN_j, \quad (1.13)$$

with $\mathbb{E}[dN_j] = \eta\gamma dt$. A crucial remark (highly relevant for interpreting measurement records) is that, ***since $\sigma_z^{(j)}$ is unitary, the jump statistics is state-independent: the records $N_j(t)$ contain “only noise” and no information about the system state, and the SME is linear in $\varrho_t^{(c)}$.*** A Kraus-operator viewpoint over a time step dt makes the structure transparent:

$$M_0 = \mathbb{I} - iH_{\text{eff}}dt, \quad M_k = \sqrt{\eta dt} L_k, \quad (1.14)$$

with non-Hermitian effective Hamiltonian

$$H_{\text{eff}} = H - \frac{i}{2}\eta \sum_k L_k^\dagger L_k, \quad (1.15)$$

while the unmonitored fraction $(1 - \eta)$ contributes the deterministic term in Eq. (1.12).

1.2.1 First jump from $|++\rangle$: projection to definite parity subspace

Since $\sigma_z^{(1)}$ and $\sigma_z^{(2)}$ commute and $(\sigma_z^{(i)})^2 = \mathbb{I}$,

$$[\sigma_z^{(1)}, \Pi_z] = \sigma_z^{(1)} \sigma_z^{(1)} \sigma_z^{(2)} - \sigma_z^{(1)} \sigma_z^{(1)} \sigma_z^{(2)} = 0, \quad (1.16)$$

$$[\sigma_z^{(2)}, \Pi_z] = \sigma_z^{(2)} \sigma_z^{(1)} \sigma_z^{(2)} - \sigma_z^{(1)} \sigma_z^{(2)} \sigma_z^{(2)} = \sigma_z^{(1)} - \sigma_z^{(1)} = 0. \quad (1.17)$$

Therefore $[\sigma_z^{(1)} \pm \sigma_z^{(2)}, \Pi_z] = 0$, and by (1.9) we have

$$[\hat{C}_\pm, \Pi_z] = 0. \quad (1.18)$$

A jump produced by $\hat{C}_\pm$ preserves the z -parity: if a pre-jump pure state $|\psi\rangle$ is a parity eigenstate, so is the post-jump normalized state : $|\psi'\rangle = \hat{C}_\pm |\psi\rangle / \|\hat{C}_\pm |\psi\rangle\|$. For pure-state jumps we use the standard update rule

$$p_k = \langle \psi | C_k^\dagger C_k | \psi \rangle dt, \quad (1.19)$$

with $k \in \{+, -\}$ and dt a short time step.

It is convenient to evaluate the *direction* of the unnormalized state $\hat{C}_\pm |++\rangle$ using the numerators of (??):

$$(\sigma_z^{(1)} + \sigma_z^{(2)}) |++\rangle = 2(|11\rangle - |00\rangle) = -2\sqrt{2} |\phi_-\rangle, \quad (1.20)$$

$$(\sigma_z^{(1)} - \sigma_z^{(2)}) |++\rangle = 2(|10\rangle - |01\rangle) = -2\sqrt{2} |\psi_-\rangle. \quad (1.21)$$

Hence the *normalized* post-jump states are

$$|++\rangle \xrightarrow{C_+} -|\phi_-\rangle, \quad |++\rangle \xrightarrow{C_-} -|\psi_-\rangle. \quad (1.22)$$

Overall minus signs are global phases and carry no physical effect; we keep them to make the subsequent alternation visually clear.

Equations (1.22) show that the first click already fixes the z -parity: C_+ lands in the even sector (Bell- ϕ states), while C_- lands in the odd sector (Bell- ψ states). From (??),

$$C_\pm^\dagger C_\pm = \frac{\gamma}{2} (S_z^{(1)} \pm S_z^{(2)})^2 = \frac{\gamma}{2} [(S_z^{(1)})^2 + (S_z^{(2)})^2 \pm 2S_z^{(1)} S_z^{(2)}]. \quad (1.23)$$

Using $S_z^2 = \frac{1}{4} \mathbb{I}$ and $S_z^{(1)} S_z^{(2)} = \frac{1}{4} \sigma_z^{(1)} \sigma_z^{(2)} = \frac{1}{4} \Pi_z$, we obtain the identity:

$$C_\pm^\dagger C_\pm = \frac{\gamma}{4} (\mathbb{I} \pm \Pi_z). \quad (1.24)$$

Therefore, for any pure state $|\psi\rangle$, the transition probability rate is:

$$p_\pm(|\psi\rangle) = \langle \psi | C_\pm^\dagger C_\pm | \psi \rangle dt = \frac{\gamma}{4} (1 \pm \langle \Pi_z \rangle_\psi) dt. \quad (1.25)$$

For the state $|++\rangle$, since $\langle \sigma_z \rangle_{|++\rangle} = 0$, we have $\langle \Pi_z \rangle_{|++\rangle} = \langle \sigma_z \rangle_{|+\rangle} \langle \sigma_z \rangle_{|+\rangle} = 0$, hence

$$p_+(|++\rangle) = p_-(|++\rangle) = \frac{\gamma}{4} dt. \quad (1.26)$$

The two channels are equiprobable on the initial state, and the total jump rate is $\frac{\gamma}{2} dt$.

For any even-parity state (eigenvalue +1 of Π_z ; in particular $|\phi_\pm\rangle$),

$$p_+(\text{even}) = \frac{\gamma}{2} dt, \quad p_-(\text{even}) = 0. \quad (1.27)$$

For any odd-parity state (eigenvalue -1; in particular $|\psi_\pm\rangle$),

$$p_-(\text{odd}) = \frac{\gamma}{2} dt, \quad p_+(\text{odd}) = 0. \quad (1.28)$$

1.2.2 Subsequent jumps: two-state cyclicity

Using again only the numerators (because normalization is restored at each jump), we find the action on Bell states:

$$\hat{C}_+ |\phi_-\rangle = -|\phi_+\rangle, \quad \hat{C}_+ |\phi_+\rangle = -|\phi_-\rangle, \quad (1.29)$$

$$\hat{C}_- |\psi_-\rangle = -|\psi_+\rangle, \quad \hat{C}_- |\psi_+\rangle = -|\psi_-\rangle. \quad (1.30)$$

After renormalization, this yields the deterministic cycles

$$-|\phi_-\rangle \xrightarrow{C_+} |\phi_+\rangle \xrightarrow{C_+} -|\phi_-\rangle \xrightarrow{C_+} \dots, \quad -|\psi_-\rangle \xrightarrow{C_-} |\psi_+\rangle \xrightarrow{C_-} -|\psi_-\rangle \xrightarrow{C_-} \dots. \quad (1.31)$$

Because of $[C_\pm, \Pi_z] = 0$, the “wrong” channel is forbidden inside each parity subspace.

1.2.3 State by state transitions - properties

Let us collect both the post-jump states and the *unconditional* jump probabilities over a time step dt :

Pre-jump state	Channel	Post-jump state & probability
$ ++\rangle$	C_+	$- \phi_-\rangle, p_+ = \frac{\gamma}{4}dt$
$ ++\rangle$	C_-	$- \psi_-\rangle, p_- = \frac{\gamma}{4}dt$
$- \phi_-\rangle$	C_+	$ \phi_+\rangle, p_+ = \frac{\gamma}{2}dt$
$ \phi_+\rangle$	C_+	$- \phi_-\rangle, p_+ = \frac{\gamma}{2}dt$
(even sector)	C_-	forbidden, $p_- = 0$
$- \psi_-\rangle$	C_-	$ \psi_+\rangle, p_- = \frac{\gamma}{2}dt$
$ \psi_+\rangle$	C_-	$- \psi_-\rangle, p_- = \frac{\gamma}{2}dt$
(odd sector)	C_+	forbidden, $p_+ = 0$

The selection rule is a direct consequence of $[C_\pm, \Pi_z] = 0$: jumps cannot mix parity sectors. Equation (1.24) makes this quantitative. Thus, once the first click has selected a parity subspace, only the corresponding channel can ever fire; conditioned on a click, the state update within that sector is deterministic as in (1.31).

Let's analyze the variances within each channel's cycle. For Bell states the correlation matrix is diagonal:

$$\langle \hat{\sigma}_\alpha^{(1)} \otimes \hat{\sigma}_\alpha^{(2)} \rangle = \begin{cases} \text{diag}(+1, -1, +1) & \text{for } |\phi_+\rangle \text{ with } \alpha = x, y, z, \\ \text{diag}(-1, +1, +1) & \text{for } |\phi_-\rangle, \\ \text{diag}(+1, +1, -1) & \text{for } |\psi_+\rangle, \\ \text{diag}(-1, -1, -1) & \text{for } |\psi_-\rangle. \end{cases}$$

Using

$$\hat{J}_\alpha^2 = \frac{1}{2} \left(\hat{\mathbb{I}} + \hat{\sigma}_\alpha^{(1)} \otimes \hat{\sigma}_\alpha^{(2)} \right), \quad \langle \hat{J}_\alpha \rangle = 0 \text{ for Bell states,}$$

one gets

$$\Delta J_\alpha^2 = \langle \hat{J}_\alpha^2 \rangle = \frac{1}{2} \left(1 + \langle \hat{\sigma}_\alpha^{(1)} \otimes \hat{\sigma}_\alpha^{(2)} \rangle \right).$$

So we summarize in a table of variances ΔJ_α^2 , for the two cycles):

state	ΔJ_x^2	ΔJ_y^2	ΔJ_z^2
$ \phi_-\rangle$	0	1	1
$ \phi_+\rangle$	1	0	1
$ \psi_-\rangle$	0	0	0
$ \psi_+\rangle$	1	1	0

1.2.4 Ensemble-averaged quantities over trajectories

For a pure two-qubit state $|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$, the concurrence is $\mathcal{C}(|\psi\rangle) = 2|ad - bc|$. For a separable state it is 0 and for any Bell states it is 1. Hence $\mathcal{C}(|++\rangle) = 0$ and $\mathcal{C}(|\phi_\pm\rangle) = \mathcal{C}(|\psi_\pm\rangle) = 1$.

Along each trajectory the concurrence is a step process: it remains 0 until the *first* click, and jumps to 1 thereafter (all subsequent clicks keep the state maximally entangled within the same parity subspace).

The waiting time to the first click is exponential with rate

$$\lambda = \text{Tr} \left[(\hat{C}_+^\dagger \hat{C}_+ + \hat{C}_-^\dagger \hat{C}_-) \cdot \rho_{++} \right] = \frac{\gamma}{2},$$

so the survival probability of no click up to time t is $P_0(t) = e^{-\lambda t} = e^{-(\gamma/2)t}$. Hence the ensemble-averaged concurrence over many trajectories is

$$\overline{\mathcal{C}}(t) = \text{prob}(\text{at least one click by } t) = 1 - e^{-(\gamma/2)t}$$

With $\gamma = 1$, $\overline{\mathcal{C}}(t) = 1 - e^{-t/2}$.

2

Appendix

2.1 Unitary invariance of Lindblad operators

Let U be a $i \times i$ unitary matrix acting on the vector of the collapsing operators $\hat{C}_i$. We want to prove that:

$$\sum_i \mathcal{D} [\hat{C}_i] = \sum_i \mathcal{D} \left[\sum_j U_{ij} \hat{C}_j \right]$$

First, we're going to explicit the dissipation superoperator in the r.h.s:

$$\sum_i \mathcal{D} \left(\sum_j U_{ij} \hat{C}_j \right) \varrho = \sum_i \left[\sum_{j,k} U_{ij} \hat{C}_j \varrho (U_{ik} \hat{C}_k)^\dagger + \frac{1}{2} \left\{ \sum_{j,k} (U_{ij} \hat{C}_j)^\dagger (U_{ik} \hat{C}_k), \varrho \right\} \right] =$$

We may bring the summation over i inside and group the terms involving the unitary matrix:

$$= \sum_{j,k} \left[\sum_i (U_{ij} U_{ki}^*) \hat{C}_j \varrho \hat{C}_k^\dagger + \frac{1}{2} \left\{ \sum_i (U_{ji}^* U_{ik}) \hat{C}_j^\dagger \hat{C}_k, \varrho \right\} \right]$$

Since $UU^\dagger = U^\dagger U = I$ we have that: $\sum_i (U_{ij} U_{ki}^*) = \sum_i (U_{ji}^* U_{ik}) = \delta_{j,k}$ so:

$$= \sum_j \hat{C}_j \varrho \hat{C}_j^\dagger + \frac{1}{2} \left\{ \hat{C}_j^\dagger \hat{C}_j, \varrho \right\} = \sum_j \mathcal{D} [\hat{C}_j] \quad \blacksquare$$

Applied in our system, the Lindblad master equation transforms as:

$$d\varrho = -i[H, \varrho] + \gamma \mathcal{D} \left[\sigma_z^{(1)} \right] \varrho + \gamma \mathcal{D} \left[\sigma_z^{(2)} \right] \varrho = -i[H, \varrho] + \gamma \mathcal{D} \left[\frac{\sigma_z^{(1)} + \sigma_z^{(2)}}{\sqrt{2}} \right] \varrho + \gamma \mathcal{D} \left[\frac{\sigma_z^{(1)} - \sigma_z^{(2)}}{\sqrt{2}} \right] \varrho$$

Last identity follows from applying the unitary matrix related to Hadamard $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

to the vector $\left(\sigma_z^{(1)}, \sigma_z^{(2)} \right)^T$